

Fake News Detection

Liantsoa Randrianasimbolarivelo

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Dataset Summary

The Fake News Classification Dataset contains over 45,000 unique English-language news articles. These articles are classified as **true (1)** or **false (0)**, providing a valuable resource for fake news detection research using Transformers models.

Exploratory Data Analysis (EDA)

Dataset Structure

- Train data: (24,353 rows, 4 columns)
- Test data: (8,117 rows, 4 columns)
- Evaluation data: (8,117 rows, 4 columns)

Data Types

Column Name	Data Type
Unnamed: 0	int64 (index)
title	object (text)
text	object (text)
label	int64 (binary classification)

Data Cleaning

- Removed the **Unnamed: 0** column (irrelevant index).
- Checked for missing values:
 - Train data: No missing values.
 - Test data: No missing values.
 - Evaluation data: No missing values.

Target Variable Distribution

- Pie chart illustrating the proportion of true vs. fake news.

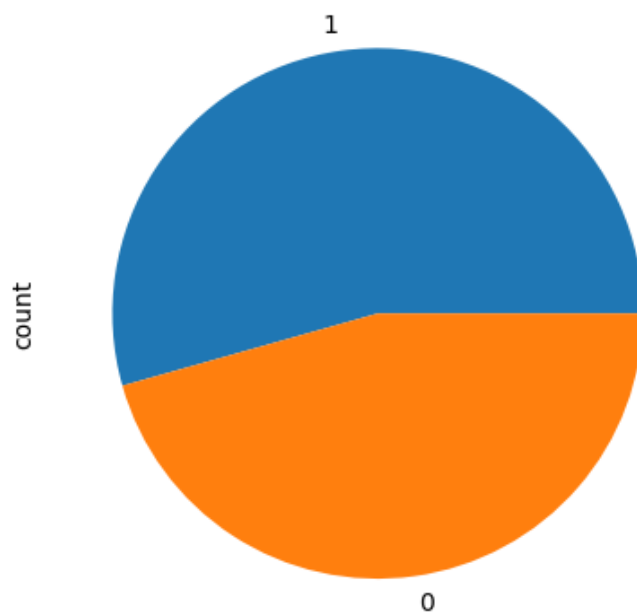


Figure 1: Proportion of true vs fake in trainset

Text Length Analysis

- Average length of news articles per class (real vs. fake).
Average length of news for label 0 : 1923.609525524444
Average length of news for label 1 : 1856.9079722180281
- Most frequent words used in true and fake news articles.

Preprocessing

- Merged `title` and `text` columns into a single `news` variable.
- Removed punctuation and special characters.
- Converted text to lowercase.
- Removed stopwords (e.g., "the", "is", "and").

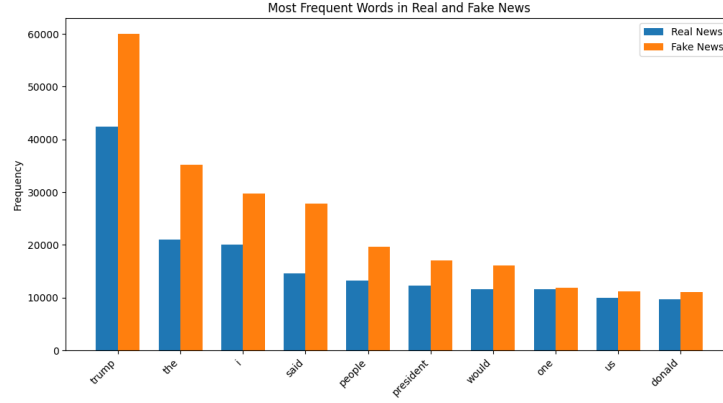


Figure 2: Most frequent words

- Tokenization and lemmatization of words.
- Applied TF-IDF transformation:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \quad (1)$$

$$\text{TF}(t, d) = \frac{\text{Number of occurrences of term } t \text{ in document } d}{\text{Total number of terms in document } d} \quad (2)$$

$$\text{IDF}(t, D) = \log \left(\frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t} \right) \quad (3)$$

Modelization

Baseline Models

- Logistic Regression:
 - Train Accuracy: 98.18%
 - Test Accuracy: 97.51%
- Decision Tree Classifier:
 - Train Accuracy: 100%
 - Test Accuracy: 97.89%
- Random Forest Classifier:
 - Train Accuracy: 100%
 - Test Accuracy: 97.37%

Final Model: Decision Tree with Hyperparameter Tuning

- Selected after grid search optimization.
- Classification report:

Class	Precision	Recall	F1-Score
Fake News (0)	0.98	0.99	0.98
Real News (1)	0.99	0.98	0.98
Accuracy		98%	

Conclusion

This study explored various methods for fake news detection. Traditional machine learning models like Logistic Regression and Decision Trees performed well, but future improvements could involve deep learning approaches like LSTM and Transformer-based models (e.g., BERT, RoBERTa). Expanding the dataset and refining preprocessing techniques may also enhance classification performance.